



Time Series Analysis

Seminar 5: Conditional volatility: ARCH and GARCH



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Acronyms used in this material

- ▣ AI: Artificial Intelligence
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ ARCH: AutoRegressive Conditional Heteroskedasticity
- ▣ ARCH-LM: ARCH Lagrange Multiplier (test)
- ▣ ARMA: AutoRegressive Moving Average (model)
- ▣ ARMA-GARCH: ARMA model for the mean with GARCH errors
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ BNR: Banca Națională a României (the National Bank of Romania)
- ▣ DM: Diebold–Mariano (test)
- ▣ EGARCH: Exponential GARCH
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ EWMA: Exponentially Weighted Moving Average
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GJR: Glosten–Jagannathan–Runkle (GARCH model)
- ▣ GJR-GARCH: Glosten–Jagannathan–Runkle GARCH
- ▣ HAC: Heteroskedasticity and Autocorrelation Consistent (standard errors)
- ▣ IGARCH: Integrated GARCH
- ▣ LR: Likelihood Ratio (test)
- ▣ MLE: Maximum Likelihood Estimation
- ▣ OLS: Ordinary Least Squares
- ▣ QLIKE: Quasi-LIKelihood (loss function)
- ▣ SE: Standard Error

Today's question and route

- ▣ **Question:** how do we detect, model, estimate and forecast a variance that changes every day?
 - ▶ this seminar comes **before** Lecture 5: the section "What you need today" gives every definition the tasks use
- ▣ Route
 - ▶ Part A: GARCH algebra, forecasts and VaR 1%, an ARCH-LM test, asymmetric news, on paper
 - ▶ Part B: ARCH effects and AR(1)-GARCH(1,1)-t, asymmetry, out-of-sample forecasts on the BET, S&P 500, EUR/RON and Bitcoin
 - ▶ Part C: an open question on the leu and an AI answer to audit
- ▣ Notebook for today: [open the seminar notebook in Google Colab](#); its first cell installs the arch package (`pip install arch`)

Exercise map

Task	Question	Type	Model
A1, A2	persistence, long-run variance, half-life and one GARCH(1,1) step	Solved, Proposed	A1
A3, A4	multi-step forecasts, 10-day volatility and VaR 1%	Solved, Proposed	A3
A5, A6	an ARCH-LM test from regression output; news impact in GJR and EGARCH	Solved, Proposed	A5, A1
B1, B2	ARCH effects and AR(1)-GARCH(1,1)-t: BET; EUR/RON and Bitcoin	Solved, Proposed	B1
B3, B4	asymmetry and the leverage effect: S&P 500; BET, EUR/RON, Bitcoin	Solved, Proposed	B3
B5, B6	GARCH against EWMA out of sample, VaR 1%: S&P 500; BET, EUR/RON	Solved, Proposed	B5
C1, C2	is the leu calmer than it used to be? what is wrong in an AI answer?	Proposed	B1, A1

- **[Solved]**: full solution in the slides and in the notebook, a model to follow; **[Proposed]**: you solve it, following the model

Data used

Series	Source	Frequency	Period
S&P 500, BET	EODHD	daily, close	2000–2026
EUR/RON	BNR reference rate	daily	2005–2026
Bitcoin	EODHD	daily, 7 days a week	2014–2026

- Daily log returns in %: $r_t = 100 (\ln P_t - \ln P_{t-1})$; weekends and repeated holiday closes are dropped (except Bitcoin)
- EUR/RON moves about 0.3% a day: the notebook estimates its models on 10 r_t and converts the results back to %
- In the notebook: `returns('bet')`, `arch_lm(x)`, `fit(r, mean='AR', dist='t')` (package `arch`); no account or key is needed

What you need today (1/4): ARCH effects

- $r_t = \mu_t + \varepsilon_t$, $\varepsilon_t = \sigma_t z_t$; z_t i.i.d., mean 0, variance 1 (the **innovations**)
 - ▶ μ_t : the **conditional mean** (an ARMA model, [Chapter 2](#)); $\sigma_t^2 = \text{Var}(r_t \mid \text{past})$: the **conditional variance**
 - ▶ daily returns are almost uncorrelated, but their squares are correlated: **volatility clustering**
- **ARCH-LM test** [Engle \(1982\)](#): regress $\hat{\varepsilon}_t^2$ on a constant and $\hat{\varepsilon}_{t-1}^2, \dots, \hat{\varepsilon}_{t-q}^2$ by OLS
 - ▶ H_0 : no ARCH effects; $\text{LM} = nR^2 \sim \chi^2(q)$; 5% values: $\chi^2(5)$ 11.07, $\chi^2(10)$ 18.31
 - ▶ the Ljung–Box $Q(m)$ of $\hat{\varepsilon}_t^2$ [Ljung and Box \(1978\)](#) is a second test with the same H_0
- **Kurtosis** $K = E[(r - \mu)^4]/\sigma^4$: 3 for the Normal distribution, larger for heavy tails

What you need today (2/4): ARCH, GARCH and EWMA

- ▣ **ARCH(1)** Engle (1982): $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2$, $\omega > 0$, $0 \leq \alpha < 1$
 - ▶ unconditional variance $\omega/(1 - \alpha)$; with Normal z_t and $3\alpha^2 < 1$: kurtosis $3(1 - \alpha^2)/(1 - 3\alpha^2)$
- ▣ **GARCH(1,1)** Bollerslev (1986): $\sigma_t^2 = \omega + \alpha \varepsilon_{t-1}^2 + \beta \sigma_{t-1}^2$, $\omega > 0$, $\alpha, \beta \geq 0$
 - ▶ **persistence** $\alpha + \beta$; **long-run variance** $\bar{\sigma}^2 = \omega/(1 - \alpha - \beta)$ if $\alpha + \beta < 1$; annual volatility $\sqrt{252 \bar{\sigma}^2}$
 - ▶ **half-life** $h_{1/2} = \ln 0.5 / \ln(\alpha + \beta)$ days: the time for half of a variance shock to disappear
- ▣ **EWMA** J.P. Morgan/Reuters (1996): $\sigma_t^2 = \lambda \sigma_{t-1}^2 + (1 - \lambda) r_{t-1}^2$ (exponential smoothing of r_t^2 , Chapter 0)
 - ▶ a GARCH with $\omega = 0$ and $\alpha + \beta = 1$: an **IGARCH** (integrated GARCH): no half-life, no long-run variance
 - ▶ **AR(1)-GARCH(1,1)**: $r_t = c + \phi r_{t-1} + \varepsilon_t$ with GARCH errors, both parts estimated together

What you need today (3/4): forecasts, VaR and estimation

▣ Forecasts made at the end of day t

► one day: $\sigma_{t+1}^2 = \omega + \alpha(r_t - \mu)^2 + \beta\sigma_t^2$; h days: $E_t[\sigma_{t+h}^2] = \bar{\sigma}^2 + (\alpha + \beta)^{h-1}(\sigma_{t+1}^2 - \bar{\sigma}^2)$

► H -day variance: $\sum_{h=1}^H E_t[\sigma_{t+h}^2] = H\bar{\sigma}^2 + (\sigma_{t+1}^2 - \bar{\sigma}^2) \frac{1 - (\alpha + \beta)^H}{1 - \alpha - \beta}$

▣ **VaR 1%** (value at risk): the loss exceeded with probability 1%: $\text{VaR}_{t+1} = -(\mu + \sigma_{t+1} q_{0.01}(z))$

► Normal: $q_{0.01} = -2.326$; standardised Student-t: $q_{0.01} = t_{\nu}^{-1}(0.01)\sqrt{(\nu - 2)/\nu}$, for example $t_5^{-1}(0.01) = -3.365$

▣ **MLE** (maximum likelihood estimation): $\hat{\theta}$ maximises $\ell(\theta) = -\frac{1}{2} \sum_t [\ln 2\pi + \ln \sigma_t^2 + (r_t - \mu_t)^2 / \sigma_t^2]$; arch reports robust SE (standard errors) [Bollerslev and Wooldridge \(1992\)](#)

► heavy-tailed innovations: standardised Student-t with ν degrees of freedom [Bollerslev \(1987\)](#); LR (likelihood ratio) test $2(\ell_1 - \ell_0) \sim \chi^2(r)$, $\chi_{0.95}^2(1) = 3.84$

What you need today (4/4): asymmetry and forecast comparison

- **GJR-GARCH** Glosten, Jagannathan and Runkle (1993): $\sigma_t^2 = \omega + (\alpha + \gamma I_{t-1})\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$, $I_{t-1} = 1$ if $\varepsilon_{t-1} < 0$
 - ▶ **leverage effect** Christie (1982): $\gamma > 0$, falls raise volatility more than rises; persistence $\alpha + \beta + \gamma/2$
 - ▶ **EGARCH** Nelson (1991): $\ln \sigma_t^2 = \omega + \alpha(|z_{t-1}| - E|z_{t-1}|) + \gamma z_{t-1} + \beta \ln \sigma_{t-1}^2$; $E|z| = 0.7979$ for Normal z ; leverage: $\gamma < 0$
 - ▶ **sign-bias test** Engle and Ng (1993): $\chi^2(3)$ on the standardised residuals $\hat{z}_t = (r_t - \hat{\mu}_t)/\hat{\sigma}_t$; a rejection means a missing asymmetry
- **Out of sample**: the forecast h_t for day t uses only data up to $t - 1$; the proxy of the true variance is r_t^2
 - ▶ **QLIKE** loss Patton (2011): $L_t = r_t^2/h_t + \ln h_t$; the smaller the mean, the better
 - ▶ **DM** (Diebold–Mariano) test Diebold and Mariano (1995): t statistic of the mean of $L_t^A - L_t^B$ with HAC (autocorrelation-robust) SE; $t < -1.96$: A is better
 - ▶ **VaR exceedance**: a day with $r_t < -\text{VaR}_t$; a correct VaR 1% is exceeded on about 1% of the days

A1: GARCH(1,1) algebra [Solved]

Task

- ▣ Daily returns in %: $\omega = 0.02$, $\alpha = 0.08$, $\beta = 0.90$, $\mu = 0$, 252 trading days per year.
- ▣ 1. Compute the persistence and the long-run variance.
- ▣ 2. Compute the annualised long-run volatility.
- ▣ 3. Compute the half-life of a variance shock.
- ▣ 4. Today $\sigma_t^2 = 1.5$ and $r_t = -3\%$; compute σ_{t+1}^2 and σ_{t+1} .
- ▣ Report: four numbers and one sentence.

Solution

- ▣ 1. $\alpha + \beta = 0.98$; $\bar{\sigma}^2 = 0.02 / (1 - 0.98) = 1.000$
- ▣ 2. $\sqrt{252 \times 1.000} = 15.87\%$ per year
- ▣ 3. $\ln 0.5 / \ln 0.98 = -0.69315 / (-0.02020) = 34.3$ days
- ▣ 4. $0.02 + 0.08 \times 9 + 0.90 \times 1.5 = 2.090$; $\sigma_{t+1} = 1.446\%$
- ▣ A fall of 3% raises the variance by 39%; half of the excess is gone after 34.3 trading days, about 7 weeks.

A2: a more reactive market [Proposed]

Solution discussed in class

Task

- ▣ $\omega = 0.05$, $\alpha = 0.12$, $\beta = 0.85$, $\mu = 0$; today $\sigma_t^2 = 2.0$ and $r_t = +1.5\%$. Model: A1.
- ▣ 1. Compute the persistence, the long-run variance and the annualised long-run volatility.
- ▣ 2. Compute the half-life.
- ▣ 3. Compute σ_{t+1}^2 .
- ▣ 4. Write EWMA with $\lambda = 0.94$ as a GARCH(1,1) and say which of the quantities in 1–2 exist for it.
- ▣ Report: four numbers and two sentences.

A3: forecasts and VaR 1% [Solved]

Task

- ▣ The model of A1, with $\sigma_{t+1}^2 = 2.090$ and Normal innovations.
- ▣ 1. Compute $E_t[\sigma_{t+5}^2]$ and $E_t[\sigma_{t+10}^2]$.
- ▣ 2. Compute the 10-day variance and volatility.
- ▣ 3. Compare with the square-root-of-time rule $\sqrt{10} \sigma_{t+1}$ and with $\sqrt{10} \bar{\sigma}^2$.
- ▣ 4. Compute the one-day and the 10-day VaR 1%.
- ▣ Report: six numbers and one sentence.

Solution

- ▣ 1. $1 + 0.98^4(2.090 - 1) = 1 + 0.9224 \times 1.09 = 2.005$;
 $1 + 0.98^9 \times 1.09 = 1.909$
- ▣ 2.
 $10 \times 1 + 1.09 \times (1 - 0.98^{10})/0.02 = 10 + 1.09 \times 9.146 = 19.97$;
 volatility 4.47%
- ▣ 3. $\sqrt{10 \times 2.090} = 4.57\%$ (too high), $\sqrt{10} = 3.16\%$ (too low)
- ▣ 4. $2.326 \times 1.446 = 3.36\%$; $2.326 \times 4.47 = 10.40\%$
- ▣ The high variance decays slowly: the 10-day risk lies between the two simple rules, closer to the first.



A4: forecasts with Student-t innovations [Proposed]

Solution discussed in class

Task

- ▣ The model of A2, with $\sigma_{t+1}^2 = 2.020$ and standardised Student-t innovations with $\nu = 5$ ($t_5^{-1}(0.01) = -3.365$). Model: A3.
- ▣ 1. Compute $E_t[\sigma_{t+10}^2]$ and the 10-day variance.
- ▣ 2. Compute the quantile $q_{0.01}$ of the standardised t.
- ▣ 3. Compute the one-day and the 10-day VaR 1% with t and with Normal innovations.
- ▣ Report: six numbers and one sentence.



A5: an ARCH-LM test from regression output [Solved]

Task

- ▣ An AR(1) is fitted to the daily returns of a stock index. The regression of $\hat{\varepsilon}_t^2$ on a constant and five lags uses $n = 1000$ observations and gives $R^2 = 0.062$; the Ljung–Box statistic of $\hat{\varepsilon}_t^2$ is $Q(10) = 95.3$.
- ▣ 1. Compute the ARCH-LM statistic and compare it with the 5% value of $\chi^2(5)$.
- ▣ 2. Compare $Q(10)$ with the 5% value of $\chi^2(10)$.
- ▣ 3. Say which model you would fit next.
- ▣ 4. For an ARCH(1) with $\omega = \alpha = 0.5$ and Normal z_t , compute the unconditional variance and the kurtosis.
- ▣ Report: two decisions, two numbers and one sentence.

Solution

- ▣ 1. $LM = 1000 \times 0.062 = 62.0 > 11.07$ ($p < 0.001$): reject H_0 , ARCH effects
- ▣ 2. $95.3 > 18.31$ ($p < 0.001$): the squares are autocorrelated; same conclusion
- ▣ 3. AR(1)-GARCH(1,1), estimated jointly; then the same tests on $\hat{z}_t$
- ▣ 4. $\bar{\sigma}^2 = 0.5/(1 - 0.5) = 1.0$; $K = 3 \times 0.75/0.25 = 9.0$: Normal innovations, yet heavy-tailed returns



A6: good and bad news [Proposed]

Solution discussed in class

Task

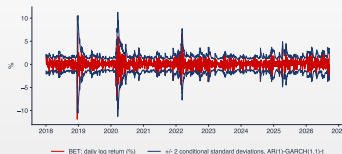
- ▣ GJR-GARCH: $\omega = 0.02$, $\alpha = 0.03$, $\gamma = 0.12$, $\beta = 0.90$; EGARCH: $\omega = 0$, $\alpha = 0.12$, $\gamma = -0.08$, $\beta = 0.98$; in both $\sigma_t^2 = 1$, $\mu = 0$. Model: A1.
- ▣ 1. Compute the GJR σ_{t+1}^2 after $\varepsilon_t = -2$ and after $\varepsilon_t = +2$, and their ratio.
- ▣ 2. Compute the GJR persistence and half-life.
- ▣ 3. Compute the EGARCH σ_{t+1}^2 for the same two shocks (use $E|z| = 0.7979$).
- ▣ Report: six numbers and one sentence.



B1: ARCH effects and AR(1)-GARCH(1,1)-t for the BET [Solved]

- ▣ **Question:** does the BET have ARCH effects, how persistent is its volatility, and where is it today relative to its long-run level?
- ▣ **Inputs:** BET closes since 2000, daily log returns in %
- ▣ **Tasks**
 1. Fit an AR(1) for the mean by OLS, then run ARCH-LM(5) and Ljung-Box $Q(10)$ on its squared residuals.
 2. Estimate AR(1)-GARCH(1,1)-t and report the parameters with robust standard errors.
 3. Compute the persistence, the half-life and the annualised long-run volatility, and compare them with the sample volatility.
 4. Check $Q(10)$ of $\hat{z}_t$ and of $\hat{z}_t^2$, and draw the returns since 2018 with $\pm 2\hat{\sigma}_t$ bands.
 5. Interpretation: is the BET volatility on the last day above or below its long-run level, and what does the model expect for the next months?
- ▣ **Report:** two test statistics, a table of six parameters, five numbers, the chart and two sentences
- ▣ **Notebook:** section B1

B1: solution [Solved]



	c	ϕ	ω	α	β	ν
estimate	0.077	0.091	0.0441	0.189	0.801	5.09
robust SE	0.010	0.013	0.0085	0.023	0.023	0.32

- AR(1) residuals: ARCH-LM = $nR^2 = 1017$ ($R^2 = 0.153$), $Q(10)$ of squares = 2496: strong ARCH effects
- $T = 6\,669$; $\alpha + \beta = 0.990$, half-life $\ln 0.5 / (-0.01028) = 67.4$ days; long-run volatility 32.8% against 22.1% in the sample; $\hat{z}_t$: $Q(10) = 28.6$ ($p < 0.001$), $\hat{z}_t^2$: 7.2 ($p = 0.705$)
- Peaks: 114% (2008), 89% (2020); last day: 26.2%; 5.1% of the days fall outside $\pm 2\hat{\sigma}_t$
- Interpretation: below (26.2% against 32.8%); the model expects the volatility to rise slowly towards the long-run level, but that level is imprecise ($1 - \alpha - \beta = 0.0102$)

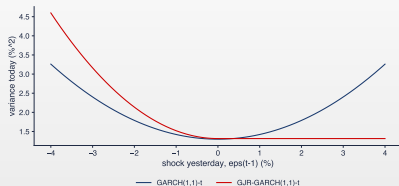
B2: EUR/RON and Bitcoin [Proposed]

- **Question:** do the EUR/RON rate and Bitcoin behave like the BET? Model: B1.
- **Inputs:** EUR/RON (BNR reference rate) since July 2005, Bitcoin since September 2014; daily log returns in %
- **Tasks**
 1. Repeat steps 1–4 of B1 for both series (EUR/RON: estimate on 10 r_t).
 2. Report $\hat{\alpha} + \hat{\beta}$ and say whether the half-life and the long-run volatility exist.
 3. Interpretation: what does $\hat{\alpha} + \hat{\beta} = 1$ mean for a forecast of the volatility one year ahead?
- **Report:** one table and two sentences
- **Notebook:** section B2

B3: the leverage effect in the S&P 500 [Solved]

- ▣ **Question:** do falls raise the volatility of the S&P 500 more than rises of the same size?
- ▣ **Inputs:** S&P 500 closes since 2000, daily log returns in %
- ▣ **Tasks**
 1. Estimate GARCH(1,1)-t and GJR-GARCH(1,1)-t; report γ with its robust t statistic.
 2. Compute the LR statistic of GJR against GARCH, its p-value and the two BIC values.
 3. Run the sign-bias test on the standardised residuals of GARCH-t.
 4. Draw both news impact curves and compute the GJR variance after shocks of -2% and $+2\%$.
 5. Interpretation: how does the leverage effect change the VaR on the day after a fall?
- ▣ **Report:** four statistics with p-values, two variances, the chart and two sentences
- ▣ **Notebook:** section B3

B3: solution [Solved]



- ▣ GJR: $\hat{\alpha} = 0.000$, $\hat{\gamma} = 0.205$ ($t = 10.0$), $\hat{\beta} = 0.881$; LR = 235.7 ($p < 0.001$); BIC 18169.0 $\rightarrow$ 17942.2
- ▣ Sign bias on the GARCH-t residuals: $\chi^2(3) = 40.7$ ($p < 0.001$), sign $t = 3.56$
- ▣ GJR variance after -2% : 2.14; after $+2\%$: 1.31; ratio 1.62
- ▣ Interpretation: only falls raise the variance ($\hat{\alpha} = 0$); after a fall the VaR is about $\sqrt{1.62} = 1.27$ times the VaR after an equal rise; a symmetric GARCH understates the risk after bad days

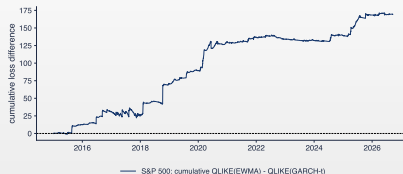
B4: asymmetry in the BET, EUR/RON and Bitcoin [Proposed]

- ▣ **Question:** is there a leverage effect in the BET, in EUR/RON and in Bitcoin? Model: B3.
- ▣ **Inputs:** BET since 2000, EUR/RON since 2005, Bitcoin since 2014; daily log returns in %
- ▣ **Tasks**
 1. Estimate GJR-GARCH(1,1)-t and EGARCH(1,1)-t; report both values of γ with robust t statistics.
 2. Compute the LR test of GJR against GARCH and the change in BIC.
 3. Interpretation: for EUR/RON a positive return means a weaker leu; what is “bad news” for this series?
- ▣ **Report:** one table and two sentences
- ▣ **Notebook:** section B4

B5: GARCH against EWMA for the S&P 500 [Solved]

- **Question:** does GARCH(1,1)-t forecast tomorrow's variance of the S&P 500 better than EWMA, and is its VaR 1% exceeded on 1% of the days?
- **Inputs:** S&P 500 since 2000; forecasts for every day since 2 January 2015
- **Tasks**
 1. Re-estimate GARCH(1,1)-t every 250 days on all data up to that day and compute the one-day forecasts h_t ; compute EWMA with $\lambda = 0.94$.
 2. Compute the mean QLIKE of both models and the Diebold–Mariano t statistic.
 3. Count the exceedances of the GARCH-t VaR 1% and of the EWMA-Normal VaR 1%, and compare them with the expected number.
 4. Draw the cumulative difference of the losses.
 5. Interpretation: which model would you give to a risk manager, and what would you still improve?
- **Report:** four numbers, two counts, the chart and two sentences
- **Notebook:** section B5

B5: solution [Solved]



- 2944 days; mean QLIKE: GARCH-t 0.761, EWMA 0.819; DM: mean difference -0.058 , $t = -3.69$ ($p < 0.001$): GARCH-t is better
- VaR 1% exceedances: GARCH-t 49 (1.66%), EWMA-Normal 65 (2.21%); expected about 29
- Interpretation: GARCH-t, but both VaR models are exceeded too often; add the leverage effect (B3) and test the number of exceedances formally ([Kupiec \(1995\)](#))

B6: GARCH against EWMA for the BET and EUR/RON [Proposed]

- ▣ **Question:** does the result of B5 hold for the BET and for EUR/RON? Model: B5.
- ▣ **Inputs:** BET since 2000, EUR/RON since 2005; forecasts for every day since January 2015
- ▣ **Tasks**
 1. Compute the out-of-sample GARCH(1,1)-t and EWMA forecasts as in B5.
 2. Compute the mean QLIKE of both models and the DM test.
 3. Find the day with the largest loss difference, and recompute the mean QLIKE without it.
 4. Count the VaR 1% exceedances of both models.
 5. Interpretation: why can the DM test be insignificant when one mean QLIKE is half the other?
- ▣ **Report:** one table and two sentences
- ▣ **Notebook:** section B6

C1: is the leu calmer than it used to be? [Proposed]

- ▣ **Question:** has the volatility of EUR/RON changed between 2005–2012, 2013–2019 and 2020–2026, and can one GARCH model describe all three periods?
- ▣ **Inputs:** EUR/RON daily log returns in % (BNR reference rate); model: B1
- ▣ **Tasks**
 1. For each period compute the annualised sample volatility and the share of days with $|r_t| > 0.5\%$.
 2. Estimate GARCH(1,1)-t on each period and report $\alpha + \beta$, α and ν .
 3. Find the largest daily move of each period and its date.
 4. Interpretation: is the change in volatility a property of the market, or of the exchange-rate policy?
- ▣ **Report:** a table and a plan for a project
- ▣ **Notebook:** section C1

C2: audit an AI answer [Proposed]

- A student asked an AI assistant to interpret a GARCH(1,1)-t fit of the BET (daily, since 2000): $\hat{\omega} = 0.0448$, $\hat{\alpha} = 0.192$, $\hat{\beta} = 0.799$, $\hat{\nu} = 5.01$. The answer:
 - (a) The half-life of a volatility shock is $\ln 0.5 / \ln \beta = 3.1$ days.
 - (b) The long-run variance is $\omega/(1 - \alpha) = 0.055$, an annual volatility of 3.7%.
 - (c) An AR(1) by OLS gives $\phi = 0.109$ with SE 0.012, $t = 8.9$: BET returns are strongly predictable.
 - (d) Ljung-Box Q(10) of the squared standardised residuals = 8.3, $p = 0.60$: the GARCH-t model is correct.
 - (e) With Student-t innovations the 1% quantile of z is closer to 0 than the Normal -2.326, so the VaR 1% is smaller.
 - (f) $\alpha + \beta = 0.991 < 1$, so the model is covariance stationary and the volatility reverts to its long-run level.
- Tasks
 - ▶ 1. For each statement, say whether it is correct; if not, give the correct statement and, where possible, the correct number from the notebook (section C2).
 - ▶ 2. Report: a list of six verdicts with one line of justification each.

Key takeaways

- ▣ Test for ARCH effects on the residuals of the mean model: ARCH-LM = nR^2 and Ljung-Box on the squares
- ▣ $\alpha + \beta$ decides the long run: half-life, long-run variance, the shape of the forecasts; $\alpha + \beta = 1$: IGARCH
- ▣ Multi-day risk is the sum of the daily variance forecasts, not $\sqrt{H}$ times today's volatility
- ▣ Heavy-tailed innovations change the VaR quantile; asymmetry changes the VaR after bad days
- ▣ Compare forecasts out of sample (QLIKE, DM) and look for single days that drive the result

After the seminar

- ▣ Lecture 5 develops each topic of today: stylised facts, ARCH and GARCH, estimation, ARMA-GARCH, asymmetry, diagnostics, forecasts, VaR 1%
- ▣ Try the [Proposed] tasks in the notebook
- ▣ C1 can grow into a team project: volatility regimes of the leu and of other currencies of the region
- ▣ Reading: [Huang and Petukhina \(2022\)](#), Ch. 6; [Tsay \(2010\)](#), Ch. 3
- ▣ **The seminar is for practice and is not graded; the solutions of [Proposed] tasks are discussed in class**

References

- Bollerslev, T. (1986). Generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 31(3), 307–327. doi:10.1016/0304-4076(86)90063-1
- Bollerslev, T. (1987). A conditionally heteroskedastic time series model for speculative prices and rates of return. *Review of Economics and Statistics*, 69(3), 542–547. doi:10.2307/1925546
- Bollerslev, T., & Wooldridge, J. M. (1992). Quasi-maximum likelihood estimation and inference in dynamic models with time-varying covariances. *Econometric Reviews*, 11(2), 143–172. doi:10.1080/07474939208800229
- Christie, A. A. (1982). The stochastic behavior of common stock variances: Value, leverage and interest rate effects. *Journal of Financial Economics*, 10(4), 407–432. doi:10.1016/0304-405X(82)90018-6
- Diebold, F. X., & Mariano, R. S. (1995). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 13(3), 253–263. doi:10.1080/07350015.1995.10524599
- Engle, R. F. (1982). Autoregressive conditional heteroscedasticity with estimates of the variance of United Kingdom inflation. *Econometrica*, 50(4), 987–1007. doi:10.2307/1912773
- Engle, R. F., & Ng, V. K. (1993). Measuring and testing the impact of news on volatility. *Journal of Finance*, 48(5), 1749–1778. doi:10.1111/j.1540-6261.1993.tb05127.x
- Glosten, L. R., Jagannathan, R., & Runkle, D. E. (1993). On the relation between the expected value and the volatility of the nominal excess return on stocks. *Journal of Finance*, 48(5), 1779–1801. doi:10.1111/j.1540-6261.1993.tb05128.x
- Huang, C., & Petukhina, A. (2022). *Applied Time Series Analysis and Forecasting with Python*. Springer. doi:10.1007/978-3-031-13584-2
- J.P. Morgan/Reuters (1996). *RiskMetrics – Technical Document* (4th ed.). Morgan Guaranty Trust Company. msci.com
- Kupiec, P. H. (1995). Techniques for verifying the accuracy of risk measurement models. *Journal of Derivatives*, 3(2), 73–84. doi:10.3905/jod.1995.407942
- Ljung, G. M., & Box, G. E. P. (1978). On a measure of lack of fit in time series models. *Biometrika*, 65(2), 297–303. doi:10.1093/biomet/65.2.297
- Nelson, D. B. (1991). Conditional heteroskedasticity in asset returns: A new approach. *Econometrica*, 59(2), 347–370. doi:10.2307/2938260
- Patton, A. J. (2011). Volatility forecast comparison using imperfect volatility proxies. *Journal of Econometrics*, 160(1), 246–256. doi:10.1016/j.jeconom.2010.03.034
- Tsay, R. S. (2010). *Analysis of Financial Time Series* (3rd ed.). Wiley. doi:10.1002/9780470644560